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CSA0911

JAVA

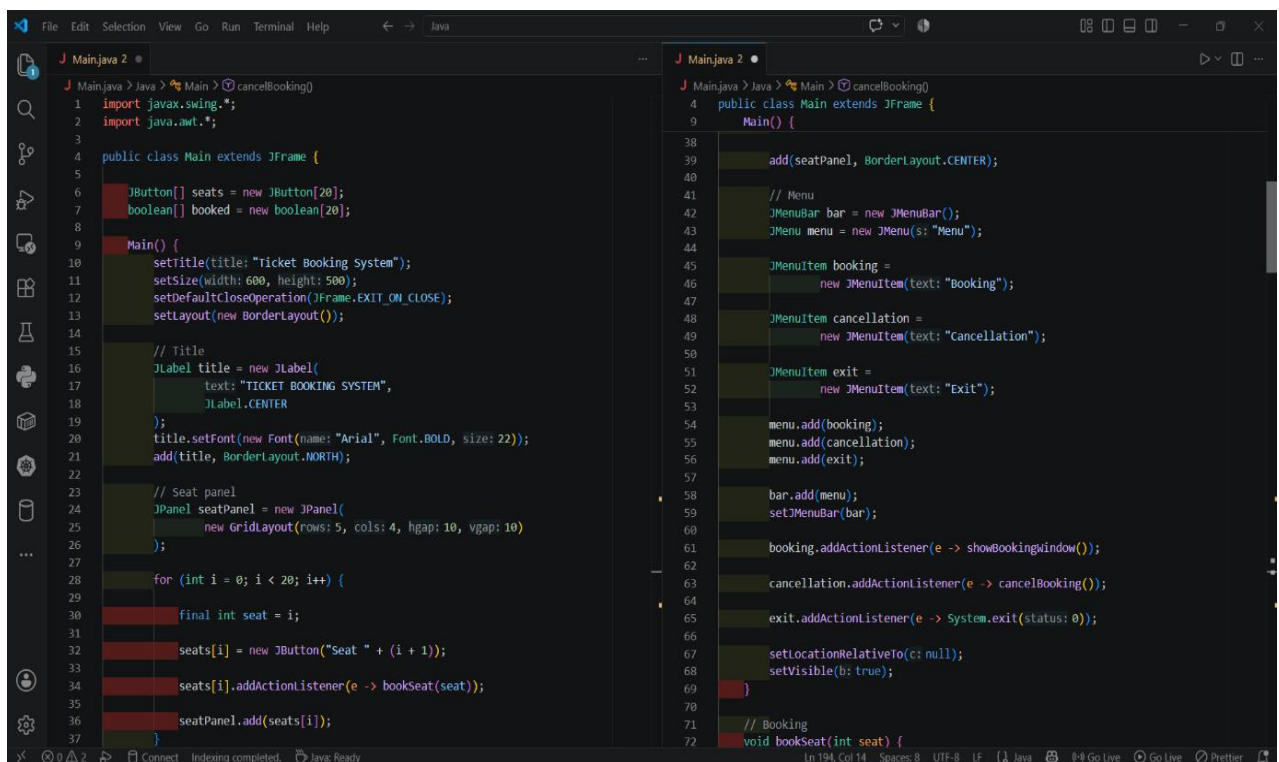
CO4 AT1 Q1

Q1 Ticket Booking System

50 marks

Develop a multi-window GUI application for ticket booking with features like seat selection, booking confirmation, and cancellation. Use Swing components such as buttons, panels, and menus along with appropriate layout managers to design the interface. Implement event handling to manage user interactions and ensure accurate seat allocation. Validate user inputs and handle exceptions during booking. Ensure smooth navigation between windows. Analyze user experience and identify areas for improving real-time interaction and responsiveness. Demonstrate the system with sample booking scenarios.

CODE:



```
1 import javax.swing.*;
2 import java.awt.*;
3
4 public class Main extends JFrame {
5
6     JButton[] seats = new JButton[20];
7     boolean[] booked = new boolean[20];
8
9     Main() {
10         setTitle(title: "Ticket Booking System");
11         setSize(width: 600, height: 500);
12         setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
13         setLayout(new BorderLayout());
14
15         // Title
16         JLabel title = new JLabel(
17             text: "TICKET BOOKING SYSTEM",
18             JLabel.CENTER
19         );
20         title.setFont(new Font(name: "Arial", Font.BOLD, size: 22));
21         add(title, BorderLayout.NORTH);
22
23         // Seat panel
24         JPanel seatPanel = new JPanel(
25             new GridLayout(rows: 5, cols: 4, hgap: 10, vgap: 10)
26         );
27
28         for (int i = 0; i < 20; i++) {
29             final int seat = i;
30
31             seats[i] = new JButton("Seat " + (i + 1));
32             seats[i].addActionListener(e -> bookSeat(seat));
33
34             seatPanel.add(seats[i]);
35         }
36     }
37
38     add(seatPanel, BorderLayout.CENTER);
39
40     // Menu
41     JMenuBar bar = new JMenuBar();
42     JMenu menu = new JMenu(s: "Menu");
43
44     JMenuItem booking =
45         new JMenuItem(text: "Booking");
46
47     JMenuItem cancellation =
48         new JMenuItem(text: "Cancellation");
49
50     JMenuItem exit =
51         new JMenuItem(text: "Exit");
52
53     menu.add(booking);
54     menu.add(cancellation);
55     menu.add(exit);
56
57     bar.add(menu);
58     setJMenuBar(bar);
59
60     booking.addActionListener(e -> showBookingWindow());
61     cancellation.addActionListener(e -> cancelBooking());
62     exit.addActionListener(e -> System.exit(status: 0));
63
64     setLocationRelativeTo(c: null);
65     setVisible(b: true);
66 }
67
68 // Booking
69 void bookSeat(int seat) {
70
71
72 }
```

```
File Edit Selection View Go Run Terminal Help Java
J Mainjava 2
J Mainjava > Java > Main > cancelBooking()
4 public class Main extends JFrame {
72 void bookSeat(int seat) {
74     if (booked[seat]) {
75
76         JOptionPane.showMessageDialog(
77             this,
78             "Seat " + (seat + 1) + " is already booked!"
79         );
80
81         return;
82     }
83
84     showBookingWindow(seat);
85 }
86
87 void showBookingWindow() {
88     showBookingWindow(-1);
89 }
90
91 void showBookingWindow(int selectedSeat) {
92
93     JTextField name =
94         new JTextField();
95
96     JComboBox<Integer> seatBox =
97         new JComboBox<>();
98
99     for (int i = 0; i < 20; i++) {
100
101         if (!booked[i]) {
102             seatBox.addItem(i + 1);
103         }
104     }
105
106     if (selectedSeat != -1) {
107         seatBox.setSelectedItem(
108             selectedSeat + 1
109         );
110     }
111
112     JPanel panel = new JPanel(
113         new GridLayout(rows: 2, cols: 2, hgap: 10, vgap: 10)
114     );
115
116     panel.add(new JLabel(text: "Passenger Name:"));
117     panel.add(name);
118
119     panel.add(new JLabel(text: "Seat Number:"));
120     panel.add(seatBox);
121
122     int result = JOptionPane.showConfirmDialog(
123         this,
124         panel,
125         title: "Booking",
126         JOptionPane.OK_CANCEL_OPTION
127     );
128
129     if (result == JOptionPane.OK_OPTION) {
130
131         String passenger =
132             name.getText().trim();
133
134         if (passenger.isEmpty()) {
135
136             JOptionPane.showMessageDialog(
137                 this,
138                 message: "Passenger name cannot be empty!"
139             );
140
141             return;
142         }
143     }
}
```

```
File Edit Selection View Go Run Terminal Help Java
J Mainjava 2
J Mainjava > Java > Main > cancelBooking()
4 public class Main extends JFrame {
91 void showBookingWindow(int selectedSeat) {
144     int seat =
145         (Integer) seatBox.getSelectedItemAt();
146
147     if (booked[seat - 1]) {
148
149         JOptionPane.showMessageDialog(
150             this,
151             message: "Seat already booked!"
152         );
153
154         return;
155     }
156
157     booked[seat - 1] = true;
158
159     seats[seat - 1].setText(text: "BOOKED");
160     seats[seat - 1].setEnabled(b: false);
161
162     JOptionPane.showMessageDialog(
163         this,
164         "Booking Confirmed!\n\n"
165         + "Passenger: " + passenger
166         + "\nSeat Number: " + seat
167     );
168 }
169
170
171 // Cancellation
172 void cancelBooking() {
173
174     String input =
175         JOptionPane.showInputDialog(
176             this,
177             message: "Enter seat number to cancel:"
178         );
179
180     if (input == null) {
181         return;
182     }
183
184     try {
185
186         int seat =
187             Integer.parseInt(input);
188
189         if (seat < 1 || seat > 20) {
190
191             throw new Exception(
192                 message: "Seat number must be between 1 and 20."
193             );
194         }
195
196         if (!booked[seat - 1]) {
197             throw new Exception(
198                 message: "This seat is not booked."
199             );
200
201         booked[seat - 1] = false;
202         seats[seat - 1].setText(
203             "Seat " + seat
204         );
205         seats[seat - 1].setEnabled(b: true);
206         JOptionPane.showMessageDialog(
207             this,
208             "Booking cancelled successfully!\n\n"
209             + "Seat: " + seat
210         );
211     } catch (NumberFormatException e) {
212         JOptionPane.showMessageDialog(
213             this,
214             message: "Enter a valid seat number."
215         );
216     }
217 }
```

```
File Edit Selection View Go Run Terminal Help
Main.java 2
Main.java > Java > Main > cancelBooking()
4 public class Main extends JFrame {
172 void cancelBooking() {
213     message: "Enter a valid seat number."
214 };
215 } catch (Exception e) {
216     JOptionPane.showMessageDialog(
217         this,
218         e.getMessage()
219     );
220 }
221 }
Run | Debug | Run main | Debug main
222 public static void main(String[] args) {
223     SwingUtilities.invokeLater(() ->
224         new Main()
225     );
226 }
227 }
```

OUTPUT:

